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NIHVA Technologies Pvt. Ltd

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1.Requirement:

Design, supply, installation and commissioning of a fully automatic Bottle Syrup filling line which is capable of 15,000 liters / day or 25,000 bottles / day @ 600ml / bottle with all accessories. Output speed is 60 Bottle/Min.

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2. Technical Inputs:

As per Attached Annexure II. and Make list.

3. Scope of Work

Scope of work shall be the complete Detailed Engineering, Procurement of Materials, Manufacturing, testing & supply of fully automatic bottle syrup filling line with required accessories.

The scope of supply includes but not limited to the followings-

1.1 Design of fully automatic bottle syrup filling line as per NIHVA specification sheet (Annexure II) & submit design & GA drawing for approval.
1.2 Procurement of all material required for fully automatic bottle syrup filling line.
1.3 Establishment of Quality Assurance Plan & submit for NIHVA approval.
1.4 Procurement & Fabrication as per finalized Quality Assurance Plan.
1.5 Trial of complete fully automatic bottle syrup filling line.
1.6 Supply of complete fully automatic bottle syrup filling line at site.
1.7 Submission of all the As-built drawings, Data sheets & QAP adherence with necessary documents, manual of system for NIHVA final records.
1.8 Submission of Project Schedule, Weekly Progress Report.

Note: Maximum utilization of modern technologies after NIHVA approval.

2. Special requirements:

Bearing:

2.1.1 All bearings and mounting should be SKF make only.
2.1.2 A common greasing position for the all bearings to be provided.

Paint

2.1.3 The paint specification shall be compatible with the external environment consisting of fine urea dust particles which is very corrosive and hygroscopic in nature.
2.1.4 Paint Ral Code require NIHVA approval.

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Design Data:

1.1.1 2D and 3D Design soft data should be share to NIHVA.
1.1.2 Detail BOM with source should be share to NIHVA.

Ducting:

1.1.3 Ducting of Fully automatic bottle syrup filling line is Hinge Type.
1.1.4 There is separate duct for electric cable and pneumatic piping
1.1.5 All electrical cable and pneumatic pipe should be proper routing with good aesthetic.

Manual:

1.1.6 Manual of fully automatic bottle filling line is submitted to NIHVA before dispatch.
1.1.7 PLC program should be submitted to NIHVA before dispatch.
1.1.8 All Spare required to fully automatic bottle syrup filling line in future shall be mention in Manual.
1.1.9 Critical Spare part List should provide separately.
1.1.10 OEM certificate should submit to NIHVA.

Safety:

1.1.1 Maximum safety Equipment consider in Line.
1.1.2 All Motor and Gear box is proper covering.
1.1.3 Pinch point in conveyor and line should be cover by metallic cover.
1.1.4 Direction, flow indication is mention in Line.
1.1.5 The vibration limit shall be as per zones A and B of ISO 10816-1
1.1.6 Noise levels produced by any rotating equipment individually and collectively shall not exceed 85 dB (A) measured at distance of 1.5 MT from the source in any direction under any load condition.
1.1.7 All the Fasteners shall be as per ASTM A 193 B7/ASTM A 194B8 with two washers.
1.1.8 The vibration limit shall be as per zones A and B of ISO 10816-1.
1.1.9 Poka-yoke should provide for Automatic bottle syrup filling line.

Motor and Gear Box:

1.1.1 Motor Insulation must be at least F class
1.1.2 Efficiency Class: IE2/IE3.
1.1.3 All Motor and Gear box is proper covering.

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PLC and HMI:

1.1.1 PLC program should be submitted to NIHVA before dispatch.
1.1.2 HMI program should design in such a way that they show indivisible station performance.
1.1.3 HMI program should design in such a way fault finding of all station is easy.
1.1.4 Integration of all station with help of PLC and HMI is at BN enterprises scope.
1.1.5 HMI screen size is 14inch

Air Unit:

1.1.1 Air preparation Unit (APU) is compulsory for conveyor line.
1.1.2 FRL (Filter Regulator and Lubricant) should provide with proper identification color mark.

Spare Part:

1.1.1 All Spare required to fully automatic bottle syrup filling line in future shall be mention in Manual.
1.1.2 Critical Spare part List should provide separately with price.

2. **Minimum Inspection and Test Requirements:**

List of Test Report		
Sr No	Document	Internal/External
1	Surface roughness Test	External
2	Welding Test	External
3	Tank leakage Test	External
4	Raw material Test Certificate	External
5	Final inspection Assembly- Visual, dimensional	Internal
6	Noise Report (less than 85db)	Internal
7	Motor warranty card	External
8	Performance Repeatability Report for each station	Internal
9	Output min 60 bottle/Min	Internal

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3. **Expected Completion Time**

The delivery of above assignment shall be in a 6 weeks' time frame from the date of award of contract. Delayed in dispatch cause to penalty charges of 0.05% for a week or 5% of overall Po valve whichever is low.

4. **Applicable Engineering Codes & Standards:**

It is the primary responsibility of vendor to design & construct all items related to this contract in accordance with the latest industry standard & codes, including all supplements and addendums issued at time of the order.

5. **Guarantee/Warrantee:**

5.1 BN Enterprises shall guarantee all components related to this contract including spare materials against faulty design, improper material of construction and poor workmanship in addition to performance guarantee. Approval or reviewing by NIHVA for design calculation and detailed shop drawing, will not in any way absolve the BN Enterprises from his responsibility. Should any repair or replacement be necessary owing to any type of failure on account of design, material and workmanship of the item, the BN Enterprises shall in view of this guarantee be bound to replace the same either in part or whole without any time & cost implication to the purchaser. Repaired or replaced part shall also be covered by same guarantee as in case of main supply.

5.2 Guarantee/warrantee shall be valid for 24 months from date of dispatch or 18 months from date of commissioning whichever is earlier.

6. **Quality Assurance & Quality Control:**

6.1 Quality Assurance (QA) shall mean the organization set up, procedures as well as test methods and facilities developed by the vendor/supplier in order to assure that all equipment / items leaving vendor's shop are of the highest possible quality that is either equal to or better than the requirements specified.

6.2 Quality Control (QC), shall mean all the tests, measurements, checks and calibration which are to be carried out at shop in order to compare the actual characteristics of the equipment/unit/system with the specified ones, along with furnishing of the relevant documentation (certificates/ records) containing the data or result of these activities.

6.3 BN Enterprises shall submit a comprehensive description (manual) of QA/QC measures contemplated by them for implementation regarding this specification. It is contractual obligation of the vendor to develop and implement adequate QA/QC systems.

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6.4 QA/QC system shall cover all products and services of the contract i.e. documentation material, shop and site fabrication, transportation and site works, including job subcontracted by the vendor.

6.5 BN enterprise shall permit the NIHVA appointed Quality personal at their premises for inspection. NIHVA shall also communicate the visit plan of their Nominated Quality cum expediter person.

7. Enclosures

7.1 Technical Spec format Annexure II

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Mandatory Make list		
Sr No	Details	Make
1	VFD	Siemens/Delta /Actech.
2	Motor	Siemens/MMC /jenet control.
3	Gear Box	Bonfiglioli mmc/ jenet coil.
4	switch	Teknic/Schneider/Siemens
5	Relay	Omron /Trinity
6	MCB/MCCB/SMPS	Siemens/L&T/omron
7	Cable	Polycab
8	Proxy Sensor	Omron/LEUZE / AUTONIX
9	Fabre Optic sensor	Genetics /Autonix.
10	Volumetric sensor	Genetics/omron
11	Vaccume Suction	
11	PLC	Siemens/Mitsubishi
12	HMI	Siemens/MITSUBISHI
13	Impeller Blower	Omron/SELECTRON Arun macnet.
14	Hot blower	AC Motor ISO 9000 (MMC MAKE)
15	Pneumatic Cylinder	~~SMC~~/ Genetics
16	Pneumatic Fittings	~~SMC~~/Genetics
17	Pneumatic Coil	~~SMC~~/ Genetics
18	Printer	M6 Handjet Expiry Date Inkjet Printer, TIJ 2.5 Thermal foaming nozzle
19	Water Pipe Line Accessories	Finolex

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ANNEXURE - II

FORMAT - TECHNICAL SPECIFICATION

For Fully automatic line.

(Capacity – 15,000 liters / day or 25,000 bottles / day @ 600ml / bottle) with all accessories

Sr.No.	Description (Column I)	Required specification (Column II)	Mention your firm Specifications without deviations for all parameters. Mention specifically if any deviations made (Column III)
1.01	***MIXING VESSEL – Working volume 3000 LTR, Gross Volume 3600LTR***		
a	GENERAL		
	Material Of Construction	SS 316 contact parts and SS 304 non contact parts.	Yes
	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	Yes
b	PROCESS DETAILS& OPERATIONAL DATA		
	Specific Gravity	0.92gms /cc	Yes
	Operating Temperature	Ambient	Yes
c	DESIGN DATA		
	Working Volume	3000 Ltr.	Yes
	Gross Volume.	3600 Ltr ~~3000 LTR.~~ 2400 LTR.	Yes
d	MATERIAL OF CONSTRUCTION		
	Shell / Bottom / Top dish	Stainless Steel 316 Grade	Yes

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e	**DRIVE AND AGITATOR DATA**			
	Type of agitator	Baby Anchor with 2 Blade PBT	Yes	✓
	Mixing speed	60 minutes for 3000 liters	Yes	✓
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	Yes	
f	**NOZZLES - TOP DISH**			
	Man Hole	450 mm with dished cover and 100 mm sight glass	Yes	✓
	Light Glass	100 MM Pad Type with FLP Vessel lamp	Yes	✓
g	**SHELL**			
	Temperature Sensor	½" BSP	Yes	✓
	Phase Inlet	40 mm with valve	Yes	✓
	Recirculation	50 mm with Valve	Yes	✓
h	**BOTTOM DISH**			
	Product Outlet	50 MM with ball valve	Yes	✓
	Filter	Suitable filter (<200 micron)	Yes	✗
i	**SURFACE FINISH**			
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	Yes	✓
	Product non-contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	Yes	✓

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1.02	***STORAGE VESSEL – Working volume 3000 LTR, Gross Volume 3300LTR***		
	MOC	SS 316 contact parts and SS 304 non contact parts.	Yes
	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	Yes
b	**Process and Operation Data**		
	Gross Volume (L)	3300 Ltrs	Yes
	Working Volume (L)	3000Ltrs	Yes
	Nozzles	Dia450mm, ManHole 50 mmInlet, 40 mm Spare25 mm outlet with valve	Yes
c	**DESIGN DATA**		
	Main Shell ID	ID Dia. 1575 mm	Yes
	Height of Shell / Cone	1500 / 500	Yes
d	**AGITATOR DATA (SUBJECT TO REVALIDATION OF DESIGN DATA)**		
	Type of agitator	Baby Anchor with 2 Blade PBT	Yes
	Agitator Entry	Top Entry	Yes
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	Yes
e	**MATERIAL OF CONSTRUCTION**		
	Product Contact Parts	Stainless Steel 316 Grade	Yes
	Product non-contact parts	Stainless Steel 304 Grade	Yes
f	**BOTTOM DISH**		Yes
	Product Outlet	50 MM with ball valve	Yes
	Filter	Suitable filter (<200 micron)	
g	**SURFACE FINISH**		Yes
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	

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	Product non-contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	Yes	✓
	Storage and Mixing Vessels.	Storage and Mixing vessel should indicate syrup Level.	Yes	✓

2.0	**LIQUID FILLING IN BOTTLE AND INNER PLUGGING AND OUTER CAPPING (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART**			
2.01	***TURN TABLE/BOTTLE FEEDER/CONTAINER LOADER***			
a	**Operation Requirements**			
	Container inputs in turn table by manually and automatically will rotate on disk of turn table and exit through outlet, during rotation of Container will guided by a SS strip, which will guide the container towards outlet path.		Yes	✓
b	**Technical Specifications**			
	Machine type	Fully automatic turn	Yes	✓
	Direction	Clockwise / Anti clock wise as per requirement	Yes	✓
	Power and Motor	0.5 to 1.0 HP/220 to 415 V, 50 to 100 cycles/ second or equivalent	0.5HP Yes	✓
	Output	100 to 120 Product /Minute (Required speed in filling line, 60BPM)	Yes	✓
c	**Salient Features** All exposed parts are of S.S. 304 to prevent rusting Required conveyor from auto turn table to air-jet cleaning machine to maintain the speed.		Yes	✓

2.02	**FULLY AUTOMATIC VERTICAL AIR-JET CLEANING MACHINE**		
a	**Technical Specifications**		
	Machine type	Fully Automatic	Yes ✓
	Out put	60 Bottles per Minute	Yes ✓
	Motor Specification	0.5 HP / 415 Volts / 50 Hz. 0.5 HP x 2 = 2 HP / 415 Volts / 50 Hz. or equivalent	Yes not Require.
	Cleaning Cycle	Positive air pressure and vacuum cleaning	Yes ✓

2.03	***FULLY AUTOMATIC VOLUMETRIC LIQUID FILLING MACHINE IN BOTTLES***	
	All parts should be made from SS-316 Materials	Yes ✓
a	**GRAND liquid Bottle size for reference**	
	Bottle size – Height 205mm, Diameter – 235mm, Neck size – 80mm.	Yes ✓
b	**Operation Requirements**	
	Nozzle should go upwards slowly from the bottom level of bottle towards neck during filling to minimize foaming adjustable nozzle is reciprocating according to filling dose.	Yes ✓
c	**Technical Specifications**	
	Machine: Fully automatic	Yes ✓
	Output: Up to 60 Bottles per minute	Yes ✓

	Filling Volume	600ml	
	Electrical Specification		Yes
	Main Machine	2.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	~~0.5 HP~~ Yes motor OR pneumatic operated.
	Conveyor	1.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	Yes
	Storage tank	SS316 with the capacity of 100 liters	Yes
	SS conveyor belt made from the SS316 (Length as per the speed of bottles filling – 60BPM) Adjustable Conveyor height as per choice to align with another line machine.		Yes

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2.04	***FULLY AUTOMATIC INNER / OUTER PLUGGING MACHINE***			
	Suitable to capping inner caps output - 60 Bottles/Minute (Bottle outer cap size: Height – above 25mm, Diameter – Above 100mm). One complete set of change parts suitable for any one size diameter of caps & bottle. Total machine construction in SS-304 wherever: Contact parts - SS-316 material.		Yes	L
a	**Operation Requirements**			
	Outer cap should be placed according to the 60 bottles filling speed.		Yes	L
b	**Technical Specifications**			
	Machine output	Up to 60 Bottles per minute	Yes	L
	MOC of machine frame	Mild Steel powder coated and outside cladding SS 304	Yes	L
	MOC of Contact Parts	Stainless Steel 316	304 Yes	L
	Machine Covers	Stainless Steel 304	Yes	L
	MOC of all vital shafts	EN 31 / Stainless Steel 304/316	304. Yes	
	MOC of Conveyor	Conveyor Slat Stainless Steel 304 Slat with Nylon Strips on Side Conveyor Shafts Mild Steel with Hard Chrome or Zinc Plating Conveyor Sprockets Mild Steel or Nylon	Yes	

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a	**Technical Specification**			
	Output	Upto30-60BottlesperMinute		Yes ✓
	No. of Sealing Head	According to the output of 60 BPM		Yes ✓
	Capping Unit	Magnetic Screw Capping Head with Torque Adjustment		Yes ✓
	Caps Feeding	Vibratory Bowl Feeder		Yes ✓
	Caps Hammering	Pneumatic System		Yes ✓
	Conveyor Length	Depending upon the product output of 60BPM	No	Yes ✓
b	**Machine's material of construction**			Yes ✓
	Machine main frame	Square Pipe, Stainless Steel 304		Yes ✓
	Machine Coverings	Complete in Stainless Steel 304		Yes ✓
	Product Contact Parts	Stainless Steel 316	304	Yes
c	**Machine's electronics & pneumatics components**			
	Capping Head Motor Make	1 H.P Standard Model		Yes ✓
	Capping Head VFD Make	1 H.P Delta		Yes ✓
d	**Machine's power & air consumption requirement**			
	Power	1 H.P / 230 V/AC or 1.5KW or equivalent		Yes ✓
	Phase	Single Phase/ Three Phase according to the requirement		Yes ✓
	Compressed Air Required	6 bar pressure, 0.5 CFM or equivalent		Yes ✓

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3.0	***LIQUID BOTTLES DRYING TUNNEL (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART*** The fully automatic drying tunnel should be designed with precision, these machines are widely recognized for their durability, design excellence and high performance.		
a	**Technical Specifications**		
	Machine Type	Fully automatic	Yes ✓
	Temp. Req.	100` To 150` Degree Centigrade in Chamber,	Yes ✓
	Heater	Heater With Heating System. Total 10 To 24 Kw	Yes ✓
	Motive Load	1.5 Hp Conveyor and 0.5 Hp Blower Fan Load	Yes ✓
b	**Material**		
	•Incoming MCB 32amp indo copper make or equivalent. •Blower Motor's are remi make or equivalent. •Blower Motor are separately controlled.		Yes ✓

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4.0	***LIQUID BOTTLES PACKING IN CARTON BOX UNIT WITH ACCESSORIES FOR INSTALLATION, PACKING, CLOSING AND STRAPPING FOR TRANSPORTATION – REQUIRED NO. 1 SET***		
	Carton Box size for reference: Length - 470mm, Width – 390mm and Height – 270mm - Yes		
a	*S-Conveyor*		
b	**Technical Specifications**		
	Output per Hour	60 to 65 Bottle/Minute	Yes
	Working Desk Height	860 To 910 mm according to the requirement	Yes
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-11 mtr/min	Yes
c	***Case Erector***		
d	**Technical Specifications**		
	Output per Hour (No. of cases)	8-10/Minute	Yes
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-10 mtr/min	Yes
	Material of construction	MS powder coated	Yes
e	***Roller Conveyor***		
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	Yes

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not Reqd.

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i	**Idler Roller Conveyor**			
	After case sealing from top of the box filled box will come and accumulate on this gravity roller conveyor. From this point filled box will pick manually		Yes	
j	***Carton box strapping machine***			
k	**Technical Specifications**			
	Output per Hour (No. of cases)	2-5/Minute or equal	Yes	
	Direction of Movement	Left to Right	Yes	
	Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement	Yes	
	Power consumption	400 watt	Yes	

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	Power consumption	0.37 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes
	Drive roller	SS-304	Yes

f	**Case Tapping Machine**		
g	**Technical Specifications**		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	Yes
	Power consumption	0.75 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes
h	**Pick and Place Unit**		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	Yes
	Power consumption	0.37 Kw according to the requirement	Yes
	Speed	0-20 mtr/min	Yes

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5.0	***FULLY AUTOMATIC BOTTLE PRINTER TO PRINT PLACE OF MANUFACTURE, DATE OF MANUFACTURE, BATCH CODE AND EXPIRY DATE***		
a	Technical specifications		
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	Yes
	Voltage	90 V to 260 V,47 Hz to 63 Hz	Yes
	Resolution	8 dots per mm	Yes
	Print area	30mm to 50mm	Yes
	Message Storage Capability	250	Yes
	Maximum Lines of Print	5, 32pixel	Yes
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	Yes

Ducting:

All Electrical cable of Bottle filling line is proper route and kept in hinge type duct.

There is separate duct for Electrical cable and Pneumatic Pipe.

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